

Assessment of Tethys

This document contains products used to assess the quality of the Tethys global DTM in this bundle. Tethys has an imaging campaign that would be classified as sufficient for SPC's purposes. See Weirich et al. (2022) for definitions of a sufficient image set. Details of the assessment data found in document/productdescription.pdf should be read before proceeding. Taken as a whole, the assessment data indicates the uncertainties given in the product description are good. Furthermore, the imaging campaign for Tethys indicates that higher resolution regional models can be generated in many locations. Most of these locations already have maplets with a higher resolution than the global DTM, though there are a few locations where where imagery supports generating even higher resolution maplets than those already present. Note that the assessment products should be used as a guide to understand the quality of the DTM.

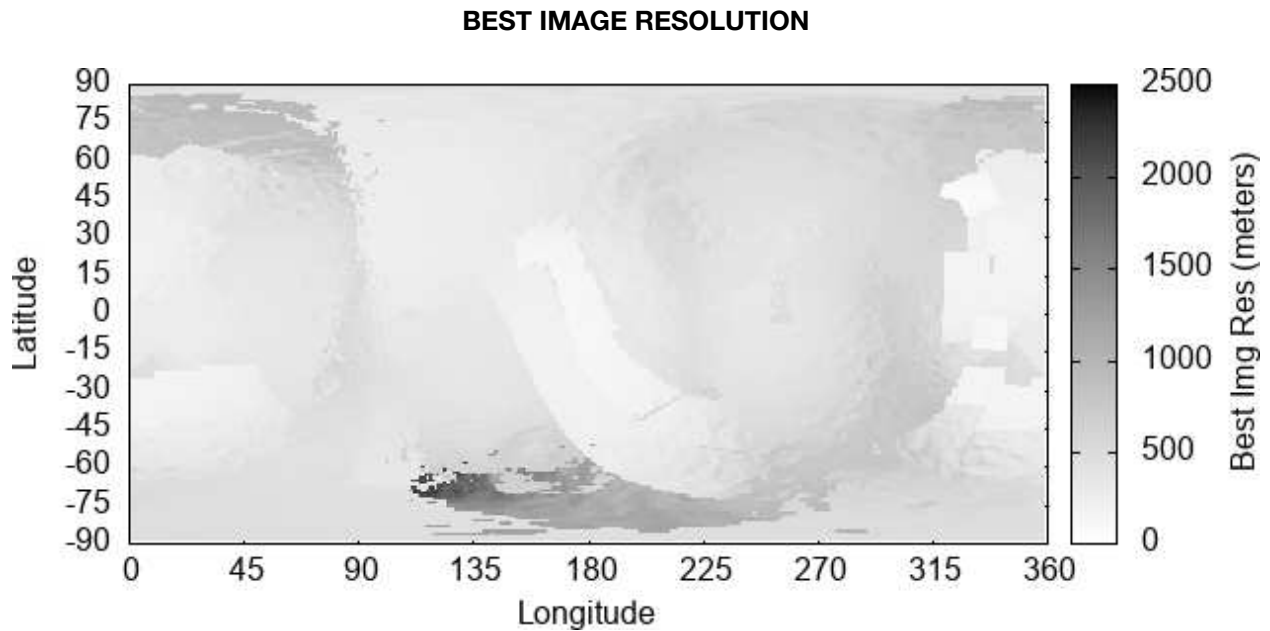
The accuracy of topography generated by SPC has been well-tested, but the relative albedo has not. If the quality of the albedo data is in doubt, a comparison with spacecraft images would be required before considering it as valid. When there is a limited imaging campaign such as Tethys's, there will be aliasing of topography into the albedo channel. Additionally, unmasked shadows can contribute to the darkening of the albedo. This darkening can be expressed as directional asymmetry in craters, especially near the poles. Where there is significant aliasing there is error in both the albedo and topography. The albedo artifacts, and thus the topography, could be improved with additional viewing geometries, but this data is not available.

Below are figures of the assessment data, along with a short exposition on the quality of each. Figures of radius and albedo are included. All figures shown here are available in the bundle as digital files that can be used for quantitative purposes. For assessment figures, the grayscale progression has been adjusted so brighter regions always represent better quality data.

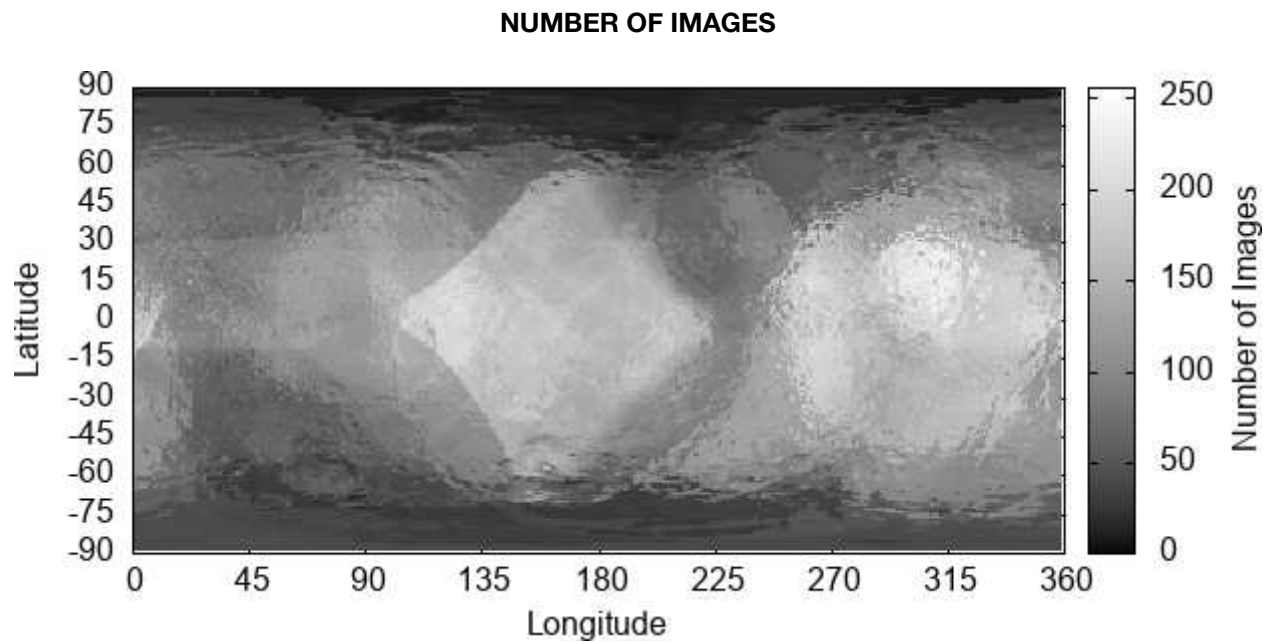
Figures of the assessment data with more detailed explanation are below.

Tethys

Spacecraft Imaging Campaign

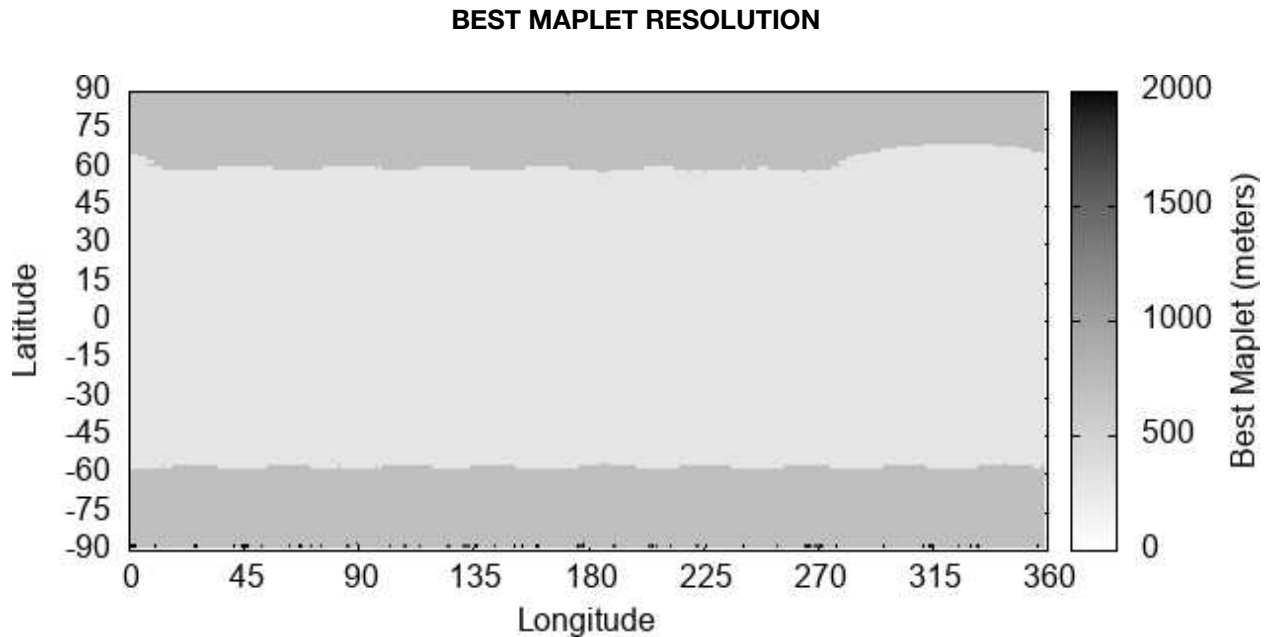


This figure presents the same information as `tethys_bestimg_c.cub`. The worst image resolution is ~ 2240 m/px and occurs near -65° latitude and 130° east longitude. The best image resolution is ~ 60 m/px and occurs near -45° latitude and 330° east longitude. The vast majority of Tethys ($\sim 99\%$) has been imaged at ~ 1500 m/px or better matching the GSD of the Q=512 global DTM. The only area where the best image resolution is worse than the Q=512 model is the dark area from about -60° latitude and 110° east longitude to -75° latitude and 155° east longitude. In this dark area the Q=512 model has an effective resolution equivalent to the best image resolution, not that of the GSD of the model. Hence the topography data of the Q=512 model is oversampled in the dark areas. The Q=128 model, with a GSD of ~ 5960 m, has a best image resolution better than the model GSD everywhere, so it does not oversample the topography data anywhere.



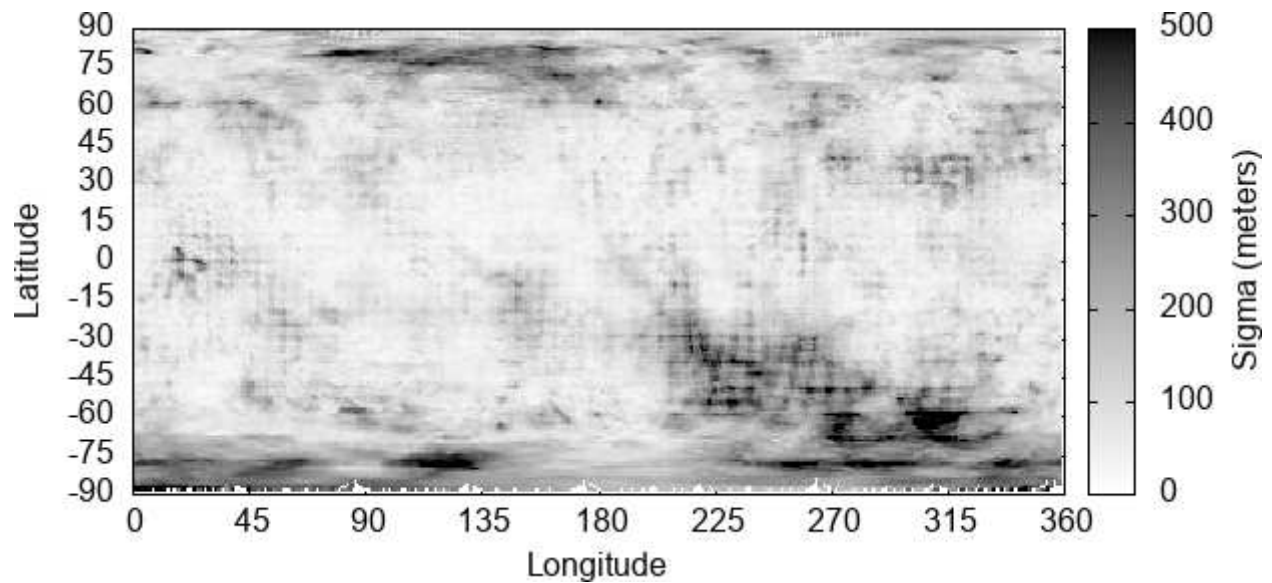
This figure presents the same information as `tethys_numimg_c.cub`. Number of images is everywhere greater than 8, and usually (i.e. ~98% of the surface) greater than 12. The number of images give confidence to the estimated radius uncertainties. Note that this data product only counts images that are better than 2800 m/px and reports pixels with more than 255 images as 255 (i.e. off scale).

Global DTM Assessment



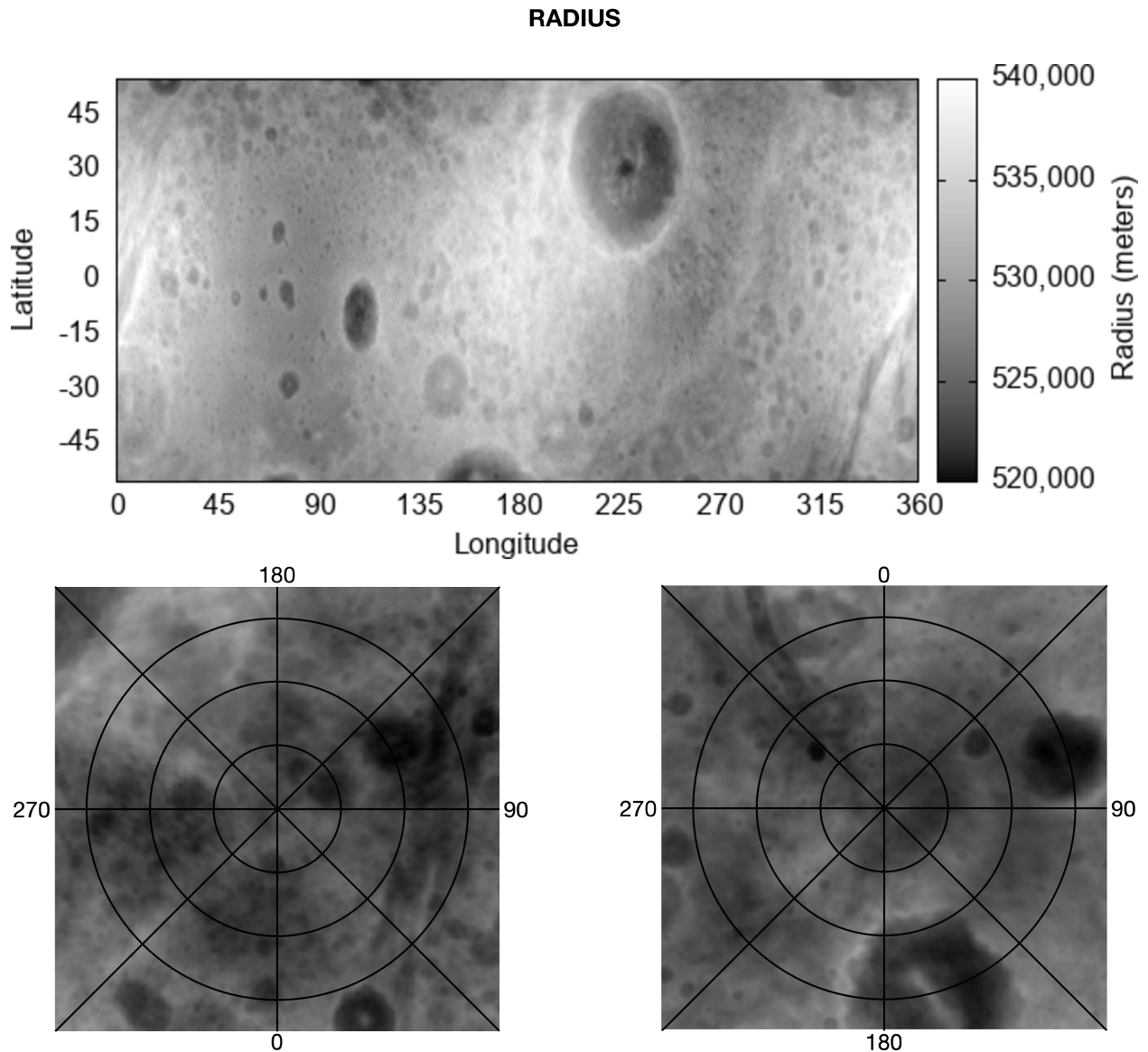
This figure presents the data in `tethys_bestmap_c.cub`. Tethys is covered by maplets with a GSD of 300 m along equatorial and mid-latitude regions, and 700 m over the whole globe. The graphic above has black pixels at the poles which indicate a higher (i.e. worse) GSD, though this is only a sampling gap. The best maplet resolution is everywhere more than ~2 times better than the Q=512 global model, indicating that the maplet resolution is not the limiting factor for uncertainty.

SIGMAS

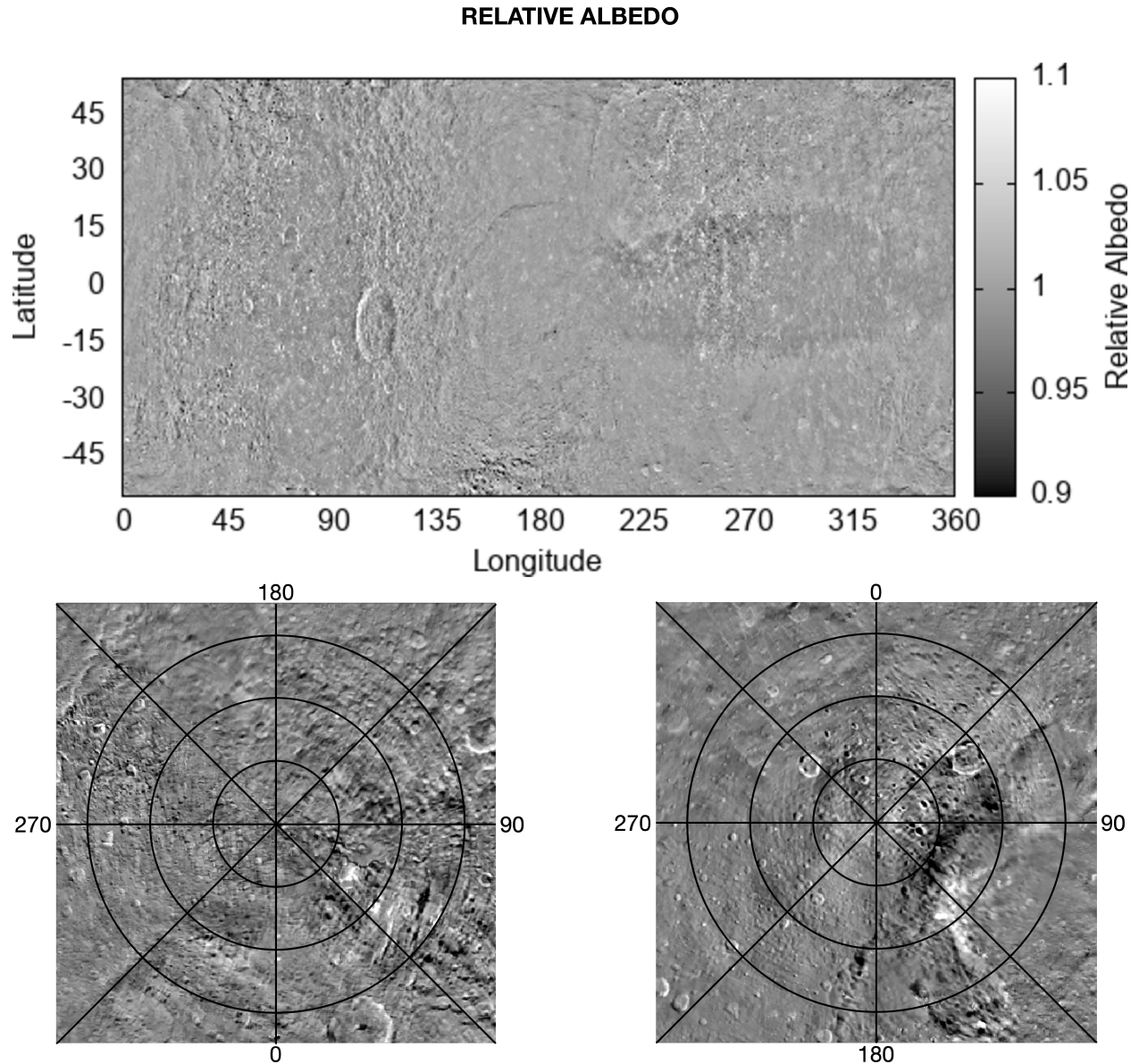


This figure presents the data in `tethys_sigmas_c.cub`. The worst sigma is ~912 m (note the color bar stops at 500 m), which is less than the GSD of the Q=512 model, and the vast majority (~99%) of the model has a sigma better than ~480 m, which is less than a third the GSD of the Q=512 model. These values indicate the maplets agree to a high confidence level. **Sigmas everywhere are good for the global model, hence the reasons for higher sigmas in various regions are outside the scope of this project.** Data gaps at the poles of this graph are due to the projection as there is complete coverage of the body.

Global DTM Values



Top: This panel presents the data in `tethys_eqradius_c.cub` and `tethys_eqradius_g.tif`.
Bottom: The left panel presents the data in `tethys_npradius_c.cub` and `tethys_npradius_g.tif`, while the right panel presents the data in `tethys_spradius_c.cub`, and `tethys_spradius_g.tif`. Circles indicate 80, 70, and 60 degrees latitude. The brightness scale is similar to the top panel.



Top: This panel represents the data in tethys_eqalbedo_c.cub and tethys_eqalbedo_g.tif.

Bottom: The left panel presents the data in tethys_npalbedo_c.cub and tethys_npalbedo_g.tif, while the right panel presents the data in tethys_spalbedo_c.cub, and tethys_spalbedo_g.tif. Circles indicate 80, 70, and 60 degrees latitude. The brightness scale is similar to the top panel.

Due to the limited imaging campaign there may be aliasing of topography into the albedo channel.